LOAN APPROVAL PREDICTION USING RANDOM FOREST

1. Problem Statement

The objective is to predict whether a loan application will be approved or rejected based on applicant details such as income, loan amount, credit history and education.

2. Dataset

Dataset: Loan Prediction Problem Dataset
Source: Kaggle
Records: 614 **Columns:** 13
Link: kaggle.com/datasets/altruistdelhite04/loan-prediction-problem-dataset

3. Data Preprocessing

The Loan_ID column was removed. Missing numerical values were replaced using the median and categorical missing values using the most frequent value. One-Hot Encoding was applied to categorical features. The dataset was split into 80% training data (491 records) and 20% testing data (123 records).

4. Model Used

A Random Forest Classifier with 100 decision trees was trained to predict loan approval (Y) or rejection (N).

5. Model Evaluation

Evaluation Metric	Result
Accuracy	82.11%
Precision	84.62%
Recall	90.59%
F1 Score	87.50%

Confusion Matrix: TN = 24, FP = 14, FN = 8, TP = 77.

6. Findings

The most important features were Credit History (21.56%), Applicant Income (20.59%) and Loan Amount (18.17%).

7. Conclusion

The Random Forest model achieved 82.11% accuracy on the test dataset. Credit history and applicant income were among the most influential features. The project demonstrates the use of machine learning for loan approval prediction. This is an academic prototype, not a system for making real loan decisions.